

ECE-UY 4563: Introduction to Machine Learning

Midterm Solutions, Fall 2024

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- Exam is closed-book, no calculators
- You are permitted two double-sided cheat sheets.
- Exam time is 80 minutes
- Write answers on the pages. If you run out of pages, please ask the instructor or course assistant for more pages
- Write your name and ID on the top right of ALL pages.

Good luck!

1. *Linear Regression.* (30 points) Consider the model:

$$\hat{y} = f(\mathbf{x}, \mathbf{w}) = \begin{cases} w_1(x_0 + x_1) + w_0 & \text{if } x_1 \leq x_0 \\ w_1(x_0 + x_1) + w_2(x_1 - x_0) + w_0 & \text{if } x_1 > x_0. \end{cases} \quad (1)$$

with input $\mathbf{x}$ and parameters $\mathbf{w} = (w_0, w_1, w_2)$.

- (a) Is $f(\mathbf{x}, \mathbf{w})$ linear in the input $\mathbf{x}$?
(b) Find three basis functions $\phi_0(\mathbf{x})$, $\phi_1(\mathbf{x})$, and $\phi_2(\mathbf{x})$ such that

$$f(\mathbf{x}, \mathbf{w}) = \sum_{j=0}^2 w_j \phi_j(\mathbf{x}).$$

- (c) Write python function `predict` that outputs a vector of n predictions `yhat` from an array `x` of shape `(n,2)` and vector of coefficients `w`.

Solution:

- (a) No. There is a discontinuity of the derivative at $x_1 = x_0$.
(b) We can take the basis functions,

$$\phi_0(\mathbf{x}) = 1, \quad \phi_1(\mathbf{x}) = x_1 + x_0, \quad \phi_2(\mathbf{x}) = \begin{cases} 0 & \text{if } x_1 \leq x_0, \\ x_1 - x_0 & \text{if } x_1 > x_0 \end{cases}$$

- (c) One solution is:

```
def predict(X, w):  
    phi1 = X[:,1] + X[:,0]  
    phi2 = (X[:,1] - X[:,0])*(X[:,1] > X[:,0])  
    yhat = w[0] + w[1]*phi1 + w[2]*phi2  
    return yhat
```

2. *Model Selection.* (35 points) A medical researcher has training and test data from two patients, patient 0 and patient 1:

- $X_{tr}[i], y_{tr}[i]$ = Training data for patient $i, i = 0, 1$
- $X_{ts}[i], y_{ts}[i]$ = Test data for patient $i, i = 0, 1$

For each i , $X_{tr}[i]$ and $X_{ts}[i]$ are arrays with one row per sample, and $y_{tr}[i]$ and $y_{ts}[i]$ are corresponding vectors of target values. She has the following functions:

```
model.fit(X,y) # Fits models on data
yhat = model.predict(X) # Predicts outputs
C = np.vstack((A,B)) # Creates an array C with A on top of B.
```

- (a) **Individual models:** Write python code such that, for each patient i , it fits a *separate* model and measures the test MSE. Store the MSE for the two patients in a vector with values `mse[0]` and `mse[1]`.
- (b) **Single model:** Write python code to fit a *single* model for both patients. Measure the test error and store in a variable `mse_single`.
- (c) Which is correct:
- (A) The individual models will have lower bias error.
- (B) The common model will have lower bias error.
- (d) Which is correct:
- (A) The individual models will have lower variance error.
- (B) The common model will have lower variance error.

Solution:

- (a) The following code will measure the MSE on each patient individually:

```
npatients = 2
mse = np.zeros(npatients)
for i in range(npatients):
    model.fit(Xtr[i], ytr[i])
    yhat = model.predict(Xts[i], yts)
    mse[i] = np.mean((yhat-yts[i])**2)
```

- (b) To fit and test a single model on all patients:

```
# Stack the training and test data
Xtrs = np.vstack((Xtr[0], Xtr[1]))
ytrs = np.vstack((ytr[0], ytr[1]))
Xtss = np.vstack((Xts[0], Xts[1]))
ytss = np.vstack((yts[0], yts[1]))

model.fit(Xtrs, ytrs)
yhat = model.predict(Xtss)
mse_single = np.mean((yhat - ytss)**2)
```

- (c) **A:** The individual model is more rich, since it allows the model to be different for the two patients. So, it will have lower bias error.
- (d) **B:** Conversely, the common model is simpler with fewer total parameters. So, it will have lower variance error

3. *Binary classification* (35 points) A researcher wishes to classify a binary variable $y \in \{0, 1\}$ from a scalar variable $x \in \{-1, 1, 3\}$. She gets training data where the number of samples for each (x, y) pair is shown in Table 1.

	$x = -1$	$x = 1$	$x = 3$
$y = 0$	65	20	75
$y = 1$	35	80	25

Table 1: Problem 3: Number of training samples for each (x, y) pair.

- (a) Based only on the samples with $x = 1$, what is a good estimate of $P_1 = P(y = 1|x = 1)$?
(b) Based only on the samples with $x \neq 1$, what is a good estimate of $P_0 = P(y = 1|x \neq 1)$?
(c) Consider a logistic model:

$$P(y = 1|x) = \frac{1}{1 + e^{-z}}, \quad z = -w(x - 1)^2 + b,$$

with parameters w and b . Given values of P_0 and P_1 find w and b such that:

$$P_1 = P(y = 1|x = 1), \quad P_0 = P(y = 1|x = 3).$$

Leave your answer in terms of P_0 and P_1 . You do not need to substitute the values from parts (a) and (b).

Solution:

- (a) We can estimate the probability as:

$$P_1 = P(y = 1|x = 1) = \frac{|\{(x, y) = (1, 1)\}|}{|\{x = 1\}|} = \frac{80}{80 + 20} = 0.8,$$

where we use the notation such as $|\{(x, y) = (-1, 1)\}|$ to denote the number of samples where $(x, y) = (-1, 1)$.

- (b) When $x \neq 1$, there are two cases in the data: $x = -1$ or $x = 3$. Adding these two events:

$$\begin{aligned} P_0 &= P(y = 1|x \neq 1) \\ &= \frac{|\{(x, y) = (-1, 1)\}| + |\{(x, y) = (3, 1)\}|}{|\{x = -1\}| + |\{x = 3\}|} \\ &= \frac{35 + 25}{(35 + 65) + (25 + 75)} = \frac{60}{100 + 100} = 0.3. \end{aligned}$$

- (c) For any x , we have:

$$P(y = 1|x) = \frac{1}{1 + e^{-z}} \implies z = \log \left[\frac{P(y = 1|x)}{1 - P(y = 1|x)} \right].$$

Since $z = -w(x - 1)^2 + b$, we have:

$$z = -w(x - 1)^2 + b = \log \left[\frac{P(y = 1|x)}{1 - P(y = 1|x)} \right].$$

At $x = 1$ we get:

$$b = \log \left[\frac{P_1}{1 - P_1} \right].$$

At $x = 3$ we get:

$$\begin{aligned} -w(2)^2 + b &= \log \left[\frac{P_0}{1 - P_0} \right] \\ \implies 4w &= \log \left[\frac{P_0}{1 - P_0} \right] - b \\ \implies w &= \frac{1}{4} \left[\log \left[\frac{P_0}{1 - P_0} \right] - \log \left[\frac{P_1}{1 - P_1} \right] \right] = \frac{1}{4} \log \left[\frac{P_0(1 - P_1)}{(1 - P_0)P_1} \right]. \end{aligned}$$

The last simplification is not necessary to receive full marks.